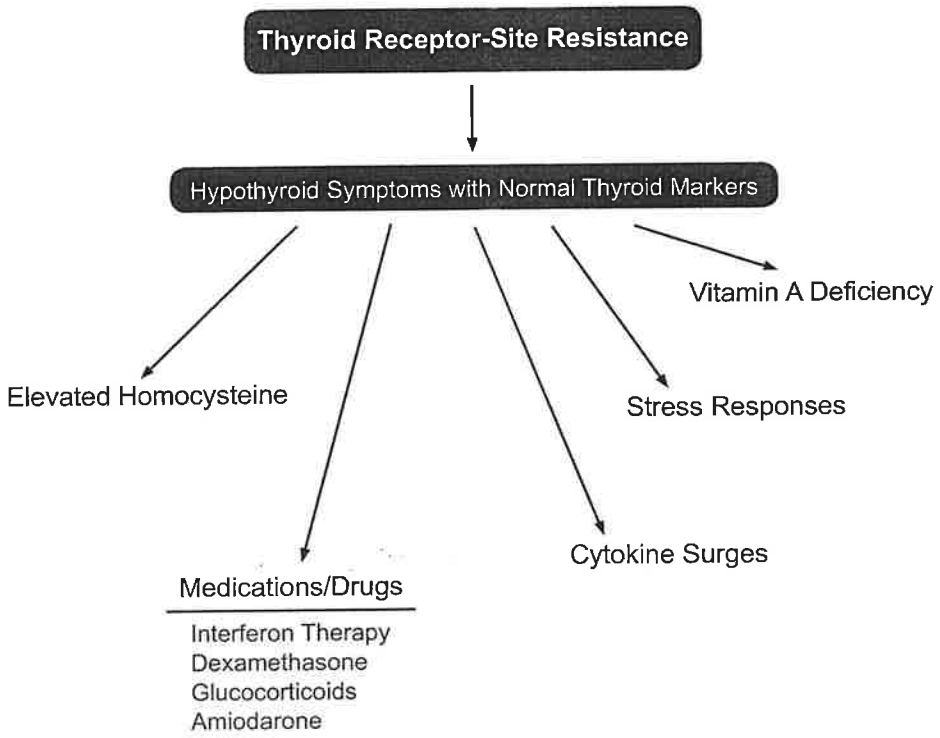

For treatment of this condition: Seek appropriate conventional medical management and drug administration by a physician.

Appendix 1 – Registered Trademark Ownership

The copyright holder of this seminar does not claim any sponsorship or approval of, or affiliation with the following individuals and entities. Each trademark is the property of its respective owner.

Abilify® is a registered trademark owned by Otsuka America Pharmaceutical, Inc.

Activella® is a registered trademark owned by Novo Nordisk A/S.

Adderall® is a registered trademark owned by Shire LLC.

Advair® is a registered trademark owned by GlaxoSmithKline LLC.

Alu-cap™ is a trademark owned by Meda Pharmaceuticals Inc.

Aludrox® is a registered trademark owned by American Home Products Corporation.

Amphojel® is a registered trademark owned by Aurium Pharma Inc.

Anadrol® is a registered trademark owned by Meda Pharmaceuticals Inc.

Androderm® is a registered trademark owned by Actavis Inc.

Androgel® is a registered trademark owned by AbbVie Inc.

Android® is a registered trademark owned by Valeant Pharmaceuticals International, Inc.

Armour® is a registered trademark owned by Forest Laboratories, Inc.

Atamet® is a registered trademark owned by Élan Pharmaceuticals PLC.

Atromid-S® is a registered trademark owned by Pfizer Inc.

Bactrim® is a registered trademark owned by Mutual Pharmaceutical Company, Inc.

Brevicon® is a registered trademark owned by Actavis Inc.

Celestone® is a registered trademark owned by Merck Sharp & Dohme Corporation.

Clomid® is a registered trademark owned by Sanofi LLC.

Colestid® is a registered trademark owned by Pfizer Inc.

Cordarone® is a registered trademark owned by Pfizer Inc.

Crestor® is a registered trademark owned by the AstraZeneca group of companies.

Cytomel® is a registered trademark owned by Pfizer Inc.

Decadron® is a registered trademark owned by Merck Sharp & Dohme Corporation.

Delatestryl® is a registered trademark owned by Endo Pharmaceuticals Inc.

Depo-Medrol® is a registered trademark owned by Pfizer Inc.

Desogen® is a registered trademark owned by Merck Sharp & Dohme Corporation.

Diabinese® is a registered trademark owned by Pfizer Inc.

Diovan® is a registered trademark owned by Novartis Pharmaceuticals Corporation.

Ecotrin® is a registered trademark owned by Prestige Brands Holdings, Inc.

Eskalith® and Eskalith® CR are registered trademarks owned by GlaxoSmithKline LLC.

Estrace® is a registered trademark owned by Actavis Inc.

Estratab® is a registered trademark owned by Solvay S.A.

Estrostep® is a registered trademark owned by Pfizer Inc.

Excedrin® is a registered trademark owned by Novartis Pharmaceuticals Corporation.

Effexor XR® is a registered trademark owned by Pfizer Inc.

Fiorinal® is a registered trademark owned by Actavis Inc.

Geref® is a registered trademark owned by EMD Serono, Inc.

Haldol® is a registered trademark owned by Janssen Pharmaceuticals, Inc.

Humatrope® is a registered trademark owned by Eli Lilly and Company.

Infergen® is a registered trademark owned by Kadmon Corporation, LLC.

Inderal® is a registered trademark owned by Akrimax Pharmaceuticals, LLC.

Intron® A is a registered trademark owned by Merck Sharp & Dohme Corporation.

Kadian® is a registered trademark owned by Actavis Inc.

Kayexalate® is a registered trademark owned by Covis Pharmaceuticals, Inc.

Lasix® is a registered trademark owned by Sanofi LLC.

Levora® is a registered trademark owned by Actavis Inc.

Levothroid® is a registered trademark owned by Forest Laboratories, Inc.

Levoxyl® is a registered trademark owned by Pfizer Inc.

Lexapro® is a registered trademark owned by Forest Laboratories, Inc.

Lipitor® is a registered trademark owned by Pfizer Inc.

Lithobid® is a registered trademark owned by ANI Pharmaceuticals, Inc.

LoCholest® is a registered trademark owned by Novartis Pharmaceuticals Corporation.

Madopa® is a registered trademark owned by Roche Products (New Zealand) Ltd.

Maalox® is a registered trademark owned by Novartis Pharmaceuticals Corporation.

Menest® is a registered trademark owned by Pfizer Inc.

Mircette® is a registered trademark owned by Teva Pharmaceuticals USA, Inc.

Modicon® is a registered trademark owned by Janssen Pharmaceuticals, Inc.

MS Contin® is a registered trademark owned by Purdue Pharma L.P.

Mylanta® is a registered trademark owned by McNeil Consumer Pharmaceuticals Co.

Nature-Throid® is a registered trademark owned by RLC Labs, Inc.

Necon® is a registered trademark owned by Actavis Inc.

Nexium® is a registered trademark owned by the AstraZeneca group of companies.

Nizoral® is a registered trademark owned by Janssen Pharmaceuticals, Inc.

Norgesic™ is a trademark owned by iNova Pharmaceuticals Pty Limited.

Nutropin® is a registered trademark owned by Genentech, Inc.

Ogen® is a registered trademark owned by Pfizer Inc.

Opadry® is a registered trademark owned by BPSI Holdings LLC.

Orinase® is a registered trademark owned by Pfizer Inc.

Ortho-Cyclen® is a registered trademark owned by Janssen Pharmaceuticals, Inc.

Ortho-Novum® is a registered trademark owned by Janssen Pharmaceuticals, Inc.

Ovcon® is a registered trademark owned by Actavis Inc.

Ovral® is a registered trademark owned by Pfizer Inc.

Pacerone® is a registered trademark owned by Upsher-Smith Laboratories, Inc.

Parcopa® is a registered trademark owned by UCB, Inc.

Paser® is a registered trademark owned by Jacobus Pharmaceutical Company, Inc.

Pepsamar® is a registered trademark owned by Sanofi LLC.

Percodan® is a registered trademark owned by Endo Pharmaceuticals Inc.

Plavix® is a registered trademark owned by Bristol-Myers Squibb Company.

Premarin® is a registered trademark owned by Pfizer Inc.

Premphase® is a registered trademark owned by Pfizer Inc.

Prempro® is a registered trademark owned by Pfizer Inc.

Prevacid® is a registered trademark owned by Novartis Consumer Health, Inc.

Prevalite® is a registered trademark owned by Upsher-Smith Laboratories, Inc.

Prolopa® is a registered trademark owned by F. Hoffmann-La Roche AG.

Protropin® is a registered trademark owned by Genentech, Inc.

Questran® Light is a registered trademark owned by Bristol-Myers Squibb Company.

Regitine® is a registered trademark owned by Novartis Pharmaceuticals Corporation.

Reglan® is a registered trademark owned by Alaven Pharmaceutical LLC.

Rifadin® is a registered trademark owned by Sanofi LLC.

Rifater® is a registered trademark owned by Sanofi LLC.

Robaxisal® is a registered trademark owned by Pfizer Inc.

Roferon®-A is a registered trademark owned by F. Hoffmann-La Roche AG.

Roxanol™ is a trademark owned by Xanodyne Pharmaceuticals, Inc.

Repronex® is a registered trademark owned by Ferring Pharmaceuticals Inc.

Saizen® is a registered trademark owned by EMD Serono, Inc.

Salofalk® is a registered trademark owned by Actavis Inc.

Sandostatin® is a registered trademark owned by Novartis Pharmaceuticals Corporation.

Septra® is a registered trademark owned by Pfizer Inc.

Sinemet® is a registered trademark owned by Merck Sharp & Dohme Corporation.

Singulair® is a registered trademark owned by Merck Sharp & Dohme Corporation.

Solu-Medrol® is a registered trademark owned by Pfizer Inc.

Soma® is a registered trademark owned by Meda Pharmaceuticals Inc.

Stalevo® is a registered trademark owned by Orion Pharma (UK) Ltd.

Synthroid® is a registered trademark owned by AbbVie Inc.

Tapazole® is a registered trademark owned by Pfizer Inc.

Testoderm® is a registered trademark owned by Johnson & Johnson.

Testred® is a registered trademark owned by Valeant Pharmaceuticals International, Inc.

Thorazine® is a registered trademark owned by GlaxoSmithKline LLC.

Thyrolar® is a registered trademark owned by Forest Laboratories, Inc.

Trecator® is a registered trademark owned by Pfizer, Inc.

Tri-Norinyl® is a registered trademark owned by Actavis Inc.

Trivora® is a registered trademark owned by Actavis Inc.

Triphasil® is a registered trademark owned by Pfizer Inc.

Unithroid® is a registered trademark owned by Jerome Stevens Pharmaceuticals, Inc.

Vagifem® is a registered trademark owned by Novo Nordisk A/S.

Westhroid™ is a trademark owned by RLC Labs, Inc.

Xanax® is a registered trademark owned by Pfizer Inc.

Zovia® is a registered trademark owned by Actavis Inc.

References

1. Betterle C, Zanchetta R. Update on autoimmune polyendocrine syndromes (APS). *Acta Biomed.* 2003 Apr ;74(1):9-33.
2. Amino N. Autoimmunity and hypothyroidism. *Baillieres Clin Endocrinol Metab.* 1988 Aug;2(3):591-617.
3. Strieder TG, Prummel MF, Tijssen JG, Endert E, Wiersinga WM. Risk factors for and prevalence of thyroid disorders in a cross-sectional study among healthy female relatives of patients with autoimmune thyroid disease. *Clin Endocrinol* (Oxf). 2003 Sep;59(3):396-401.
4. Brent GA. The molecular basis of thyroid hormone action. *N Engl J Med.* 1994;331:847-853.
5. Tsai MJ, O'Malley BW. Molecular mechanisms of action of steroid/thyroid receptor superfamily members. *Annu Rev Biochem.* 1994;63:451-486.
6. Zhang J, Lazar MA. The mechanism of action of thyroid hormones. *Annu Rev Physiol* 2000;62:439-466.
7. Hermus RM, Sweep CG, van der Meer MJ, et al. Continuous infusion of interleukin-1 beta induces a nonthyroidal illness syndrome in the rat. *Endocrinology.* 1992 Nov131(:5):2139-2146.
8. Rettori V, Jurcovicoval J, McCann SM. Central action of interleukin-1 in altering the release of TSH, growth hormone, and prolactin in the male rat. *J Neurosci Res.* 1987;18(1):179-183.
9. Bartalena L, Grasso L, Brogioni S, et al: Interleukin-6 on pituitary-thyroid-axis in the rat. *Eur J Endocrinol.* 1994;131:302-306.
10. Lyson K, McCann SM. The effect of interleukin-6 on pituitary hormone release in vivo and vitro. *Neuroendocrinology.* 1991;54:262-266.
11. Ozawa M, Sato K, Han DC, Kawakami M, Tsushima T, Shizume K. Effects of tumor necrosis factor-alpha/cachectin on thyroid hormone metabolism in mice. *Endocrinology.* 1988 Sep;123(3):1461-1467.
12. Rettori V, Milenkovic L, Beutler BA, McCann SM. Hypothalamic action of cachectin to alter pituitary hormone release. *Brain Res Bull.* 1989 Dec;23(6):471-475.
13. Pang XP, Hershman JM, Mirell CJ, Pekary AE. Impairment of hypothalamic-pituitary-thyroid function in rats treated with human recombinant tumor necrosis factor-alpha (cachetin). *Endocrinology.* 1989 Jul;125(1):76-84.
14. Ongphiphadhanakul B, Fang SL, Tang KT, Patwardhan NA, Braverman LE. Tumor necrosis factor-alpha decreases thyrotropin-induced 5'-deiodinase activity in FRTL-5 thyroid cells. *Eur J Endocrinol.* 1994 May;130(5):502-507.
15. Pekary AE, Berg L, Santini F, Chopra I, Hershman JM. Cytokines modulate type I iodothyronine deiodinase mRNA levels and enzyme activity in FRTL-5 rat thyroid cells. *Moll Cell Endocrinol.* 1994 May;101(1-2):R31-R35.
16. Yu J, Koenig RJ. Regulation of hepatocyte thyroxine 5'-deiodinase by T3 and nuclear receptor coactivators as a model of the sick euthyroid syndrome. *J Biol Chem.* 2000 Dec 8;275(49):38296-38301.
17. Molnár I, Balázs C, Szegedi G, Sipka S. Inhibition of type 2,5'-deiodinase by tumor necrosis factor alpha, interleukin-6 and interferon gamma in human thyroid tissue. *Immunol Lett.* 2002 Jan 1;80(1):3-7.
18. Nagaya T, Fujieda M, Otsuka G, Yang JP, Okamoto T, Seo H. A potential role of activated NF-kappa B in the pathogenesis of euthyroid sick syndrome. *J Clin Invest.* 2000 Aug;106(3):393-402.
19. Bartalena L, Brogioni S, Grasso L, Velluzzi F, Martino E. Relationship of the increased serum interleukin-6 concentration to changes of thyroid function in nonthyroidal illness. *J Endocrinol Invest.* 1994 Apr;17(4):269-274.
20. Corssmit EP, Heyligenberg R, Endert E, Sauerwein HP, Romijn JA. Acute effects of interferon-alpha administration on thyroid hormone metabolism in healthy men. *J Clin Endocrinol Metab.* 1995 Nov;80(11):3140-3144.
21. van der Poll T, Romijn JA, Wiersinga WM, Sauerwein HP. Tumor necrosis factor: a putative mediator of the sick euthyroid syndrome in man. *J Clin Endocrinol Metab.* 1990 Dec;71(6):1567-1572.
22. Stouthard JM, van der Poll T, Endert E, et al. Effects of acute and chronic interleukin-6 administration on thyroid hormone metabolism in humans. *J Clin Endocrinol Metab.* 1994 Nov;79(5):1342-1346.
23. Rittig MG, Kaufmann A, Robins A, et al. Smooth and rough lipopolysaccharide phenotypes of Brucella induce different intracellular trafficking and cytokine/chemokine release in human monocytes. *J Leukoc Biol.* 2003 Dec;74(6):1045-1055.
24. van der Poll T, Endert E, Coyle SM, Agosti JM, Lowry SF. Neutralization of TNF does not influence endotoxininduced changes in thyroid hormone metabolism in humans. *Am J Physiol.* 1999 Feb;276(2 Pt 2):R357-R362.
25. van der Poll T, Van Zee KJ, Endert E, et al. Interleukin-1 receptor blockade does not affect endotoxin-induced changes in plasma thyroid hormone and thyrotropin concentrations in man. *J Clin Endocrinol Metab.* 1995 Apr;80(4):1341-1346.
26. Boelen A, Kwakkel J, Platvoet-ter Schiphorst M, et al. Interleukin-18, a proinflammatory cytokine, contributes to the pathogenesis of non-thyroidal illness mainly via the central part of the hypothalamus-pituitary-thyroid axis. *Eur J Endocrinol.* 2004 Oct;151(4):497-502.
27. Beigneux AP, Moser AH, Shigenaga JK, Grunfeld C, Feingold KR. Sick euthyroid syndrome is associated with decreased TR expression and DNA binding in mouse liver. *Am J Physiol Endocrinol Metab.* 2003 Jan;284(1):E228-E236.
28. Lu B, Moser AH, Shigenga JK, Feingold KR, Grunfeld C. Type II nuclear hormone receptors, coactivator, and target gene repression in adipose tissue in the acute-phase response. *J Lipid Res.* 2006 Oct;47(10):2179-2190.
29. Kimura H, Caturegli P. Chemokine orchestration of autoimmune thyroiditis. *Thyroid.* 2007 Oct;17(10):1005-1011.
30. Yamazaki K, Suzuki K, Yamada E, et al. Suppression of iodide uptake and thyroid hormone synthesis with stimulation of the type I interferon system by double-stranded ribonucleic acid in cultured human thyroid follicles. *Endocrinology.* 2007 Jul;148(7):3226-3235.
31. Kwakkel J, Wiersinga WM, Boelen A. Interleukin-1beta modulates endogenous thyroid hormone receptor alpha gene transcription in liver cells. *J Endocrinol.* 2007 Aug;194(2):257-265.
32. Klecha AJ, Genaro AM, Gorelik G, et al. Integrative study of hypothalamus-pituitary-thyroid-immune system interaction: thyroid hormone-mediated modulation of lymphocyte activity through the protein kinase C signaling pathway. *J Endocrinol.* 2006 Apr;189(1):45-55.
33. Barreiro Arcos ML, Gorelik G, Klecha A, Genaro AM, Cremaschi GA. Thyroid hormones increase inducible nitric oxide synthase gene expression downstream from PKC-zeta in murine tumor T lymphocytes. *Am J Physiol Cell Physiol.* 2006 Aug;291(2):C327-336.
34. Mascanfroni I, Montesinos Mdel M, Susperreguy S, et al. Control of dendritic cell maturation and function by triiodothyronine. *FASEB J.* 2008 Apr;22(4):1032-1042.
35. Hodkinson CF, Simpson EE, Beattie JH, et al. Preliminary evidence of immune function modulation by thyroid hormones in healthy men and women aged 55-70 years. *J Endocrinol.* 2009 Jul;202(1):55-63.
36. Yao C, Zhang J, Wang L, Guo Y, Tian Z. Inhibitory effects of thyroxine on cytokine production by T cells in mice. *Int Immunopharmacol.* 2007 Dec 15;7(13);1747-1754.
37. Koyuncu A, Aydintu S, Koçak S, et al. Effect of thyroid hormones on stress ulcer formation. *ANZ J Surg.* 2002 Sep;72(9):672-675.
38. Furr MO, Murray MJ, Ferguson DC. The effects of stress on gastric ulceration, T3, T4, reverse T3 and cortisol in neonatal foals. *Equine Vet J.* 1992 Jan;24(1):37-40.
39. Wang J, Klein JR. Hormone regulation of murine T cells: potent tissue-specific immunosuppressive effects of thyroxine targeted to gut T cells. *Int Immunol.* 1996 Feb;8(2):231-235.
40. Money SR, Cheron RG, Jaffe BM, Zinner MJ. The effects of thyroid hormones on the formation of stress ulcers in the rat. *J Surg Res.* 1986 Feb;40(2):176-180.
41. Rosolovski AP. The role of catecholamines in the mechanisms of thyroid gland influence on the development of "stress" and "steroid" stomach ulcers. *Probl Endokrinol* (Mosk). 1976 Jan-Feb;22(1):64-67.
42. Aksoy DY, Kerimoglu U, Okur H, et al. Effects of prophylactic thyroid hormone replacement in euthyroid Hashimoto's thyroiditis. *Endocr J.* 2005 Jun;52(3):337-343.
43. Karanikas G, John P, Wahl K, Schuetz M, Dudczak R, Willheim M. T-lymphocyte cytokine production patterns in nonimmune severe hypothyroid state and after thyroid hormone replacement therapy. *Thyroid.* 2004 Jul;14(7):488-492.
44. Mihara S, Suzuki N, Wakisaka S, et al. Effects of thyroid hormones on apoptotic cell death of human lymphocytes. *J Clin Endocrinol Metab.* 1999 Apr;84(4):1378-1385.
45. Legrand J. Hormones thyroidiennes et maturation du syste`me nerveux. *J Physiol (Paris). 78 :603-652.*
46. Porterfield SP, Hendrich CE. The role of thyroid hormones in prenatal and neonatal neurological development—current perspectives. *Endocr Rev.* 1993 Feb;14(1):94-106.

47. Chamak B, Dobbertin A, Mallat M. Immunohistochemical detection of thrombospondin in microglia in the developing rat brain. *Neuroscience*. 1995 Nov;69(1):177-187.

48. Elkabes S, DiCicco-Bloom EM, Black IB. Brain microglia/macrophages express neurotrophins that selectively regulate microglial proliferation and function. *J Neurosci*. 1996 Apr 15;16(8):2508-2521.

49. Frade JM, Barde YA. Microglia-derived nerve growth factor causes cell death in the developing retina. *Neuron*. 1998 Jan;20(1):35-41.

50. Lima FR, Gervais A, Colin C, Izembart M, Neto VM, Mallat M. Regulation of microglial development: a novel role for thyroid hormone. *J Neurosci*. 2001 Mar 15;21(6):2028-2038.

51. Puymirat J. Thyroid receptors in the rat brain. *Prog Neurobiol*. 1992 Sep;39(3):281-294.

52. Lima RS, Trentin AG, Rosenthal D, Chagas C, Moura Neto V. Thyroid hormone induces protein secretion and morphological changes in astroglial cells with an increase in expression of glial fibrillary acidic protein. *J Endocrinol*. 1997;154:167-175.

53. Mallat M, Calvo CF, Dobbertin A. Migration and proliferation of mononuclear phagocytes in the central nervous system. *Adv Exp Med Biol*. 1997;429:99-108.

54. Dobbertin A, Schmid P, Gelman M, Glowinski J, Mallat M. Neurons promote macrophage proliferation by producing transforming growth factor-beta2. *J Neurosci*. 1997 Jul 15;17(14):5305-5315.

55. Bernal J. Thyroid hormones and brain development. *Vitam Horm*. 2005;71:95-122

56. Calzà L, Aloe L, Giardino L. Thyroid hormone-induced plasticity in the adult rat brain. *Brain Res Bull*. 1997;44(4):549-557.

57. Stipcevic T, Kusacic-Kuna S, Dezeljin M, et al. Platelet serotonin concentration and monoamine oxidase activity in hypothyroid patients. *Horm Res*. 2009;71(4):207-212.

58. Iosifescu DV, Bolo NR, Nierenberg AA, Jensen JE, Fava M, Renshaw PF. Brain bioenergetics and response to triiodothyronine augmentation in major depressive disorder. Biol Psychiatry. 2008 Jun 15;63(12):1127-1134.

59. Kulikov AV, Zubkov EA. Chronic thyroxine treatment activates the 5-HT2A serotonin receptor in the mouse brain. *Neurosci Lett*. 2007 Apr 18;416(3):307-309.

60. Abraham G, Milev R, Stuart Laswon J T3 augmentation of SSRI resistant depression. *J Affect Disord*. 2006 Apr;91(2-3):211-215.

61. Gitlin M, Altshuler LL, Frye MA, et al Peripheral thyroid hormones and response to selective serotonin reuptake inhibitors. *J Psychiatry Neurosci*. 2004 Sep;29(5):383-386.

62. Sullo A, Brizzi G, Maffulli N Serotonin effect on deiodinating activity in the rat. *Can J Physiol Pharmacol*. 2003 Jul;81(7):747-751.

63. Bauer M, Heinz A, Whybrow PC Thyroid hormones, serotonin and mood: of synergy and significance in the adult brain. *Mol Psychiatry*. 2002;7(2):140-156.

64. Kulikov AV, Jeanningro R. The effects of hypothyroidism on 5-HT1A and 5-HT2A receptors and the serotonin transporter protein in the rat brain. *Neurosci Behav Physiol*. 2001 Jul-Aug;31(4):445-449.

65. Kulikov A, Moreau X, Jeanningros R. Effects of experimental hypothyroidism on 5-HT1A, 5-HT2A receptors, 5-HT uptake sites and tryptophan hydroxylase activity in mature rat brain1. *Neuroendocrinology*. 1999 Jun;69(6):453-459.

66. Tejani-Butt SM, Yang J, Kaviani A. Time course of altered thyroid states on 5-HT1A receptors and 5-HT uptake sites in rat brain: an autoradiographic analysis. *Neuroendocrinology*. 1993 Jun;57(6):1011-1118.

67. Toivonen M, Tuomainen P, Männistö PT. Effects of hypothalamic paraventricular nucleus lesion on the cold-stimulated TSH responses to 5-HT in male rats. *Acta Physiol Scand*. 1990 May;139(1):233-240.

68. Soldin OP, Aschner M. Effects of manganese on thyroid hormone homeostasis: potential links. *Neurotoxicology*. 2007 Sep;28(5):951-956.

69. McGrath PA, Bourke JR, Huxham GJ, Manley SW. Electrical responses of cultured thyroid cells to serotonin. *J Endocrinol*. 1985 Dec;107(3):397-401.

70. Maayan ML, Sellitto RV, Volpert EM. Dopamine and L-dopa: inhibition of thyrotropin-stimulated thyroidal thyroxine release. *Endocrinology*. 1986 Feb;118(2):632-636.

71. Foord SM, Peters JR, Dieguez C, Shewring G, Hall R, Scanlon MF. Thyrotropin regulates thyrotroph responsiveness to dopamine in vitro. *Endocrinology*. 1986 Apr;118(4):1319-1326.

72. Garbutt JC, Loosen PT, Glenn M. Lack of effect of dopamine receptor blockade on the TSH response to TRH in borderline personality disorder. *Psychiatry Res*. 1987 Aug;21(4):307-311.

73. Cooper DS, Klibanski A, Ridgway EC. Dopaminergic modulation of TSH and its subunits: in vivo and in vitro studies. *Clin Endocrinol* (Oxf). 1983 Mar;18(3):265-275.

74. Strawn JR, Ekhator NN, D'Souza BB, Geracioti TD Jr. Pituitary-thyroid state correlates with central dopaminergic and serotonergic activity in healthy humans. Neuropsychobiology. 2004;49(2):84-87.

75. Carageorgiou H, Pantos C, Zarros A, et al. Changes in acetylcholinesterase, Na+,K+-ATPase, and Mg2+-ATPase activities in the frontal cortex and the hippocampus of hyper- and hypothyroid adult rats. *Metabolism*. 2007 Aug;56(8):1104-1110.

76. Carageorgiou H, Pantos C, Zarros A, et al. Effects of hyper- and hypothyroidism on acetylcholinesterase, (Na(+), K (+))- and Mg (2+)-ATPase activities of adult rat hypothalamus and cerebellum. Metab Brain Dis. 2007 Mar;22(1):31-38.

77. Smith JW, Evans AT, Costall B, Smythe JW. Thyroid hormones, brain function and cognition: a brief review. *Neurosci Biobehav Rev*. 2002 Jan;26(1):45-60.

78. Sarkar PK, Ray AK. Involvement of L-triiodothyronine in acetylcholine metabolism in adult rat cerebrocortical synaptosomes. *Horm Metab Res*. 2001 May;33(5):270-275.

79. Ahmed MT, Sinha AK, Pickard MR, Kim KD, Ekins RP. Hypothyroidism in the adult rat causes brain region-specific biochemical dysfunction. *J Endocrinol*. 1993 Aug;138(2):299-305.

80. Landa ME, González Burgos G, Cardinali DP. In vitro effect of thyroxine on cholinergic neurotransmission in rat sympathetic superior cervical ganglion. *Neuroendocrinology*. 1991 Dec;54(6):552-558.

81. Wiens SC, Trudeau VL. Thyroid hormone and gamma-aminobutyric acid (GABA) interactions in neuroendocrine systems. *Comp Biochem Physiol A Mol Integr Physiol*. 2006 Jul;144(3):332-344.

82. Ortiz-Butron R, Pacheco-Rosado J, Hernández-Garcia A, Briones-Velasco M, Rocha L. Mild thyroid hormones deficiency modifies benzodiazepine and mu-opioid receptor binding in rats. *Neuropharmacology*. 2003 Jan;44(1):111-116.

83. Go T, Ito M, Okuno T, Mikawa H. Effect of thyroid hormones on benzodiazepine receptors in neuron-enriched primary cultures. *J Neurochem*. 1988 Nov;51(5):1497-1500.

84. Mason GA, Walker CH, Prange AJ Jr, Bondy SC. GABA uptake is inhibited by thyroid hormones: implications for depression. *Psychoneuroendocrinology*. 1987;12(1):53-59.

85. Vallortigara J, Alfos S, Micheau J, Higueret P, Enderlin V. T3 administration in adult hypothyroid mice modulates expression of proteins involved in striatal synaptic plasticity and improves motor behavior. *Neurobiol Dis*. 2008 Sep;31(3):378-385.

86. Peterson AL, Gilman TL, Banks ML, Sprague JE. Hypothyroidism alters striatal dopamine release mediated by 3,4-methylenedioxymethamphetamine (MDMA, ecstasy). *Synapse*. 2006 Apr;59(5):317-319.

87. Del Compare JA, Aguirre JA, Ibarra FR, Barontini M, Armando I. Effects of thyroid hormone on the renal dopaminergic system. *Endocrine*. 2001 Aug;15(3):297-303.

88. Oh-Nishi A, Saji M, Furudate SI, Suzuki N. Dopamine D(2)-like receptor function is converted from excitatory to inhibitory by thyroxine in the developmental hippocampus. *J Neuroendocrinol*. 2005 Dec;17(12):836-845.

89. Sjöberg S, Eriksson M, Nordin C. L-thyroxine treatment and neurotransmitter levels in the cerebrospinal fluid of hypothyroid patients: a pilot study. *Eur J Endocrinol*. 1998 Nov;139(5):493-497.

90. Guerin V, Guyot C, Hartemann P. Dysthyroidism and Parkinson's disease. *Ann Endocrinol* (Paris). 1990;51(1):43-45.

91. Drucker D, Josse R. Inappropriate TSH secretion with abnormal thyrotroph sensitivity to dopamine. *Clin Invest Med*. 1985;8(2):117-120.

92. Marchesi C, Chiodera P, De Risio C, et al. Dopaminergic control of TSH secretion in endogenous depression. *Psychiatry Res*. 1988 Sep;25(3):277-282.

93. Dieguez C, Peters JR, Page MD, John R, Hall R, Scanlon MF. Thyroid function in patients with hyperprolactinaemia: relationship to dopaminergic inhibition of TSH release. *Clin Endocrinol* (Oxf). 1986 Oct;25(4):435-440.

94. Grubb MR, Chakeres D, Malarkey WB. Patients with primary hypothyroidism presenting as prolactinomas. *Am J Med*. 1987 Oct;83(4):765-769.

95. Sidneva LN, Adamskaia EI. The relationship between changes in the concentration of catecholamines in the hypothalamus and the level of thyroid hormones in the body. *Probl Endokrinol* (Mosk). 1975 Nov-Dec;21(6):84-88.

96. Sharma Y, Bajpai A, Mittal S, Kabra M, Menon PS. Ovarian cysts in young girls with hypothyroidism: follow-up and effect of treatment. *J Pediatr Endocrinol Metab*. 2006 Jul;19(7):895-900.

97. Wu X, Zhang Z, Su Y. Functional states of pituitary-ovary, adrenal and thyroid axes in women with polycystic ovarian syndrome. *Zhonghua Fu Chan Ke Za Zhi*. 1998 Mar;33(3):153-156.

98. Hatsuta M, Abe K, Tamura K, et al. Effects of hypothyroidism on the estrous cycle and reproductive hormones in mature female rat. *Eur J Pharmacol*. 2004 Feb 23;486(3):343-348.

99. Antony FF, Aruldhas MM, Udhayakumar RC, Maran RR, Govindarajulu P. Inhibition of Leydig cell activity in vivo and in vitro in hypothyroid rats. *J Endocrinol*. 1995 Feb;144(2):293-300.

100. Maran RR. Thyroid hormones: their role in testicular steroidogenesis. *Arch Androl*. 2003 Sep-Oct;49(5):375-388.

101. Valenti S, Guido R, Fazzuoli L, Barreca A, Giusti M, Giordano G. Decreased steroidogenesis and cAMP production in vitro by leydig cells isolated from rats made hypothyroid during adulthood. *Int J Androl*. 1997 Oct;20(5):279-286.

102. Kumar A, Mohanty BP, Rani L. Secretion of testicular steroids and gonadotrophins in hypothyroidism. *Andrologia*. 2007 Dec;39(6):253-260.

103. Andó S, Panno ML, Beraldi E, et al. Influence of hypothyroidism on in-vitro testicular steroidogenesis in adult rats. Exp Clin Endocrinol. 1990 Nov;96(2):149-156.

104. Meikle AW. The interrelationships between thyroid dysfunction and hypogonadism in men and boys. *Thyroid*. 2004;14 Suppl 1:S17-S25.

105. Ylöstalo P, Kujala P, Kontula K. Amenorrhea with low normal thyroid function and thyroxine treatment. *Int J Gynaecol Obstet*. 1980;18(3):176-180.

106. Bruni JF, Marshall S, Dibbet JA, Meites J. Effects of hyper- and hypothyroidism on serum LH and FSH levels in intact and gonadectomized male and female rats. *Endocrinology*. 1975 Sep;97(3):558-563.

107. Maruo T, Katayama K, Barnea ER, Mochizuki M. A role for thyroid hormone in the induction of ovulation and corpus luteum function. Horm Res. 1992;37 Suppl 1:12-18.

108. Akande EO. Plasma concentration of gonadotrophins, oestrogen and progesterone in hypothyroid women. *Br J Obstet Gynaecol*. 1975 Jul;82(7):552-556.

109. Tohei A. Studies on the functional relationship between thyroid, adrenal and gonadal hormones. *J Reprod Dev*. 2004 Feb;50(1):9-20.

110. Tolino A, Nicotra M, Romano L, Petrone A, Langella L. Subclinical hypothyroidism and hyperprolactinemia. *Acta Eur Fertil*. 1991 Sep-Oct;22(5):275-277.

111. Jiang JY, Imai Y, Umezu M, Sato E. Characteristics of infertility in female hypothyroid (hyt) mice. *Reproduction*. 2001 Nov;122(5):695-700.

112. Bispink L, Brändle W, Lindner C, Bettendorf G. Preclinical hypothyroidism and disorders of ovarian function. *Geburtshilfe Frauenheilkd*. 1989 Oct;49(10):881-888.

113. Mattheij JA, Swarts JJ, Lokerse P, van Kampen JT, Van der Heide D. Effect of hypothyroidism on the pituitary-gonadal axis in the adult female rat. *J Endocrinol*. 1995 Jul;146(1):87-94.

114. Müller MJ, Paschen U, Seitz HJ. Thyroid hormone regulation of glucose homeostasis in the miniature pig. *Endocrinology*. 1983 Jun;112(6):2025-2031

115. Kitamura S, Jinno N, Suzuki T, et al. Thyroid hormone-like and estrogenic activity of hydroxylated PCBs in cell culture. *Toxicology*. 2005 Mar 30;208(3):377-387.

116. Lin SY, Wang YY, Liu PH, Lai WA, Sheu WH. Lower serum free thyroxine levels are associated with metabolic syndrome in a Chinese population. *Metabolism*. 2005 Nov;54(11):1524-1528.

117. Purandare A, Co Ng L, Godil M, Ahnn SH, Wilson TA. Effect of hypothyroidism and its treatment on the IGF system in infants and children. *J Pediatr Endocrinol Metab*. 2003 Jan;16(1):35-42.

118. Sapronov NS, Fedotova YO. The effect of L-tryptophan on conditioned reflex learning and behavior in rats with experimental pathology of the thyroid gland. *Neurosci Behav Physiol*. 2002 May-Jun;32(3):237-241.

119. Foltyn W, Nowakowska-Zajdel E, Danikiewicz A, Brodziak A. Hypothalamic-pituitary-thyroid axis in depression. *Psychiatr Pol*. 2002;36(2):281-292.

120. Mani S. Signalling mechanisms in progesterone-neurotransmitter interactions. *J Mol Endocrinol*. 2003 Apr;30(2):127-137.

121. LoPresti JS, Nicoloff JT. Thyroid Response to Critical Illness. In: Ober KP, ed. *Endocrinology of Critical Disease*. Totowa, NJ: Humana Press; 1997:155-174.

122. Stratakis CA, Chrousos GP. Neuroendocrinology and pathophysiology of the stress system. *Ann N Y Acad Sci*. 1995 Dec 29;771:1-18.

123. Stockigt JR. Update on the Sick Euthyroid Syndrome. In: Baverman LE, ed. *Diseases of the Thyroid*. Totowa, NJ: Humana Press; 1997:49-68.

124. Bisschop PH, Toorians AW, Endert E, Wiersinga WM, Gooren LJ, Fliers E. The effects of sex-steroid administration on the pituitary-thyroid axis in transsexuals. *Eur J Endocrinol*. 2006 Jul;155(1):11-16.

125. Knopp RH, Bergelin RO, Wahl PW, Walden CE, Chapman MB. Clinical chemistry alterations in pregnancy and oral contraceptive use. *Obstet Gynecol*. 1985 Nov;66(5):682-690.

126. Steingold KA, Matt DW, DeZiegler D, Sealey JE, Fratkin M, Reznikov S. Comparison of transdermal to oral estradiol administration on hormonal and hepatic parameters in women with premature ovarian failure. *J Clin Endocrinol Metab*. 1991 Aug;73(2):275-280.

127. Geola FL, Frumar AM, Tataryn IV, et al. Biological effects of various doses of conjugated equine estrogens in postmenopausal women. *J Clin Endocrinol Metab*. 1980 Sep;51(3):620-625.

128. Ben-Rafael Z, Struass JF 3rd, Arendash-Durand B, Mastroianni L Jr, Flickinger GL. Changes in thyroid function tests and sex hormone binding globulin associated with treatment by gonadotropin. *Fertil Steril*. 1987 Aug;48(2):318-320.

129. Deyssig R, Weissel M. Ingestion of androgenic-anabolic steroids induces mild thyroidal impairment in male body builders. *J Clin Endocrinol Metab*. 1993 Apr;76(4):1069-1071.

130. Malarkey WB, Strauss RH, Leizman DJ, Liggett M, Demers LM. Endocrine effects in female weight lifters who self-administer testosterone and anabolic steroids. *Am J Obstet Gynecol*. 1991 Nov;165(5 Pt 1):1385-1390.

131. Graham RL, Gambrell RD Jr. Changes in thyroid function tests during danazol therapy. *Obstet Gynecol*. 1980 Mar;55(3):395-397.

132. Williams GR, Franklyn JA, Neuberger JM, Sheppard MC. Thyroid hormone receptor expression in the "sick euthyroid" syndrome. *Lancet*. 1989 Dec 23-30;2(8678-8679):1477-1481.

133. Riedel W, Layka H, Neeck G. Secretory pattern of GH, TSH, thyroid hormones, ACTH, cortisol, FSH, and LH in patients with fibromyalgia syndrome following systemic injection of the relevant hypothalamic-releasing hormones. *Z Rheumatol*. 1998;57 Suppl 2:81-87.

134. Sterling K. Thyroid hormone action at the cell level (first of two parts). *N Engl J Med*. 1979 Jan 18;300(3):117-123.

135. Licata A. Bone density vs bone quality: what's a clinician to do? *Cleve Clin J Med*. 2009 Jun;76(6):331-336.

136. Lakatos P. Thyroid hormones: beneficial or deleterious for bone? *Calcif Tissue Int*. 2003 Sep;73(3):205-209.

137. Kosińska A, Syrenicz A, Syrenicz M, Kosiński B, Miazgowski T, Garanty-Bogacka B. The influence of treatment with substitutive or suppressive doses of thyroxine on biochemical bone turnover markers. *Ann Acad Med Stetin*. 2005;51(2):97-104.

138. Yuan PQ, Yang H. Hypothyroidism increases Fos immunoreactivity in cholinergic neurons of brain medullary dorsal vagal complex in rats. *Am J Physiol Endocrinol Metab*. 2005 Nov;289(5):E892-E899.

139. Pustorino S, Foti M, Calipari G, et al. Thyroid-intestinal motility interactions summary. *Minerva Gastroenterol Dietol*. 2004 Dec;50(4):305-315.

140. Eastwood GI, Braverman LE, White EM, Vander Salm TJ. Reversal of lower esophageal sphincter hypotension and esophageal aperistalsis after treatment for hypothyroidism. *J Clin Gastroenterol*. 1982 Aug;4(4):307-310.

141. Green JR. Thyroid function in chronic liver disease. *Z Gastroenterol*. 1979 Jul;17(7):447-451.

142. Schlienger JL, Hasselman M, Imler M. Alterations in thyroid hormones and thyrotropin response to TRH in cirrhotic patients with or without hyperammonemia (author's transl). *Ann Endocrinol (Paris)*. 1978;39(1):75-76.

143. Krawczyńska H, Sadowska H, Malunowicz E, Obrowska D, Kowalska H, Krawczyński J. Diminished TSH response to TRH stimulation in patients with hepatic cirrhosis despite of subnormal serum T3 levels. *Mater Med Pol*. 1979 Oct-Dec;11(4):367-372.

144. Napoli R, Guardasole V, Angelini V, et al. Acute effects of triiodothyronine on endothelial function in human subjects. *J Clin Endocrinol Metab*. 2007 Jan;92(1):250-254.

145. Taddei S, Caraccio N, Virdis A, et al. Impaired endothelium-dependent vasodilatation in subclinical hypothyroidism: beneficial effect of levothyroxine therapy. *J Clin Endocrinol Metab*. 2003 Aug;88(8):3731-3737.

146. Saha B, Maity C. Alteration of serum enzymes in primary hypothyroidism. *Clin Chem Lab Med*. 2002 Jun;40(6):609-611.

147. Laukkarinen J, Kiudelis G, Lempinen M, et al. Increased prevalence of subclinical hypothyroidism in common bile duct stone patients. *J Clin Endocrinol Metab*. 2007 Nov;92(11):4260-4264.

148. Laukkarinen J, Sand J, Saaristo R, et al. Is bile flow reduced in patients with hypothyroidism? *Surgery*. 2003 Mar;133(3):288-293.

149. Inkinen J, Sand J, Nordback I. Association between common bile duct stones and treated hypothyroidism. *Hepatogastroenterology*. 2000 Jul-Aug;47(34):919-921.

150. Müller MJ, Burger AG, Ferrannini E, Jequier E, Acheson KJ. Glucoregulatory function of thyroid hormones: role of pancreatic hormones. *Am J Physiol*. 1989 Jan;256(1 Pt 1):E101-E110.

151. Oge A, Sozmen E, Karaoglu AO. Effect of thyroid function on LDL oxidation in hypothyroidism and hyperthyroidism. *Endocr Res*. 2004 Aug;30(3):481-489.

152. Fommei E, Iervasi G. The role of thyroid hormone in blood pressure homeostasis: evidence from short-term hypothyroidism in humans. *J Clin Endocrinol Metab*. 2002 May;87(5):1996-2000.

153. Maralcan G, Erkol H, Erkol Z, Yanar F, Plevin R. The effects of low thyroid hormone levels on the formation of stress gastritis: an experimental study on the rats. *Ulus Travma Acil Cerrahi Derg*. 2008 Apr;14(2):96-102.

154. Silva JE. The thermogenic effect of thyroid hormone and its clinical implications. *Ann Intern Med*. 2003 Aug 5;139(3):205-213.

155. Mano T, Sakamoto H, Fujita K, et al. Effects of thyroid hormone on catecholamine and its metabolite concentrations in rat cardiac muscle and cerebral cortex. *Thyroid*. 1998 Apr;8(4):353-358.

156. Boado RJ, Romeo HE, Chuluyan HE, Cageao L, Cardinali DP, Zaninovich AA. Evidence suggesting that the sympathetic nervous system mediates thyroidal depression in turpentine-induced nonthyroidal illness syndrome. *Neuroendocrinology*. 1991 Apr;53(4):360-364.

157. Görges R, Saller B, Eising EG, Quadbeck B, Mann K, Bockisch A. Surveillance of TSH-suppressive levothyroxine treatment in thyroid cancer patients: TRH testing versus basal TSH determination by a third generation assay. *Exp Clin Endocrinol Diabetes*. 2002 Oct;110(7):355-360.

158. Abel ED, Ahima RS, Boers ME, Elmquist JK, Wondisford FE. Critical role for thyroid hormone receptor beta2 in the regulation of paraventricular thyrotropin-releasing hormone neurons. *J Clin Invest*. 2001 Apr;107(8):1017-1023.

159. Hinkle PM, Goh KB. Regulation of thyrotropin-releasing hormone receptors and responses by L-triiodothyronine in dispersed rat pituitary cell cultures. *Endocrinology*. 1982 May;110(5):1725-1731.

160. Azizi F, Vagenakis AG, Ingbar SH, Braverman LE. The time course of changes in TRH responsiveness in man following a single dose of liothyronine. *Metabolism*. 1975 Jun;24(6):691-694.

161. Yuan PQ, Yang H. Localization of thyroid hormone receptor beta2 in the ventral medullary neurons that synthesize thyrotropin-releasing hormone. *Brain Res*. 2000 Jun 16;868(1):22-30.

162. Raggenbass M, Vozzi C, Tribollet E, Dubois-Dauphin M, Dreifuss JJ. Thyrotropin-releasing hormone causes direct excitation of dorsal vagal and solitary tract neurones in rat brainstem slices. *Brain Res*. 1990 Oct 15;530(1):85-90.

163. Taché Y, Yang H, Yoneda M. Vagal regulation of gastric function involves thyrotropin-releasing hormone in the medullary raphe nuclei and dorsal vagal complex. *Digestion*. 1993;54(2):65-72.

164. Taché Y, Vale W, Rivier J, Brown M. Brain regulation of gastric acid secretion in rats by neurogastrointestinal peptides. *Peptides*. 1981;2 Suppl 2:51-55.

165. Saperas E, Mourelle M, Santos J, Moncada S, Malagelada JR. Central vagal activation by an analogue of TRH stimulates gastric nitric oxide release in rats. *Am J Physiol*. 1995 Jun;268(6 Pt 1):G895-G899.

166. Yoneda M, Taché Y. Vagal regulation of gastric prostaglandin E2 release by central TRH in rats. *Am J Physiol*. 1993 Feb;264(2 Pt 1):G231-G236.

167. Taché Y, Yoneda M. Central action of TRH to induce vagally mediated gastric cytoprotection and ulcer formation in rats. *J Clin Gastroenterol*. 1993;17 Suppl 1:S58-S63.

168. Santos J, Saperas E, Mourelle M, Antolín M, Malagelada JR. Regulation of intestinal mast cells and luminal protein release by cerebral thyrotropin-releasing hormone in rats. *Gastroenterology*. 1996 Dec;111(6):1465-1473.

169. Taché Y, Yang H, Miampamba M, Martinez V, Yuan PQ. Role of brainstem TRH/TRH-R1 receptors in the vagal gastric cholinergic response to various stimuli including sham-feeding. *Auton Neurosci*. 2006 Apr 30;125(1-2):42-52.

170. Yang H, Taché Y, Ohning G, Go VL. Activation of raphe pallidus neurons increases insulin through medullary thyrotropin-releasing hormone (TRH)-vagal pathways. Pancreas. 2002 Oct;25(3):301-307.

171. Messmer B, Zimmerman FG, Lenz HJ. Regulation of exocrine pancreatic secretion by cerebral TRH and CGRP: role of VIP, muscarinic, and adrenergic pathways. *Am J Physiol*. 1993 Feb;264(2 Pt 1):G237-G242.

172. Goto M, Yoneda M, Nakamura K, Terano A, Haneda M. Effect of central thyrotropin-releasing hormone on pancreatic blood flow in rats. *Regul Pept*. 2004 Sep 15;121(1-3):57-63.

173. Tamori K, Yoneda M, Nakamura K, Makino I. Effect of intracisternal thyrotropin-releasing hormone on hepatic blood flow in rats. *Am J Physiol*. 1998 Feb;274(2 Pt 1):G277-G282.

174. Sato Y, Yoneda M, Nakamura K, Makino I, Terano A. Protective effect of central thyrotropin-releasing hormone on carbon tetrachloride-induced acute hepatocellular necrosis in rats. *J Hepatol*. 2003 Jul;39(1):47-54.

175. Yoneda M, Tomari K, Sato Y, et al. Central thyrotropin-releasing hormone stimulates hepatic DNA synthesis in rats. *Hepatology*. 1997 Nov;26(5):1203-1208.

176. Lenz HK, Zimmerman FG, Messmer B. Regulation of canine gallbladder motility by brain peptides. *Gastroenterology*. 1993 Jun;104(6):1678-1685.

177. Tamori K, Yoneda M, Yokohama S, et al. Role of calcitonin gene-related peptide and capsaicin-sensitive afferents in central thyrotropin-releasing hormone-induced hepatic hyperemia. *Eur J Pharmacol*. 1999 Sep 3;380(1):31-35.

178. Napoli R, Biondi B, Guardasole V, et al. Enhancement of vascular endothelial function by recombinant human thyrotropin. *J Clin Endocrinol Metab*. 2008 May;93(5):1959-1963.

179. Zachariassen KE, Flaten TP. Is fluoride-induced hyperthyroidism a cause of psychosis among East African immigrants to Scandinavia? *Med Hypotheses*. 2009 May;72(5):501-503.

180. Jooste PL, Weight MJ, Kriek JA, Louw AJ. Endemic goitre in the absence of iodine deficiency in schoolchildren of the Northern Cape Province of South Africa. *Eur J Clin Nutr*. 1999 Jan;53(1):8-12.

181. Conaway CC, Getahun SM, Liebes LL, et al. Disposition of glucosinolates and sulforaphane in humans after ingestion of steamed and fresh broccoli. *Nutr Cancer*. 2000;38(2):168-178.

182. Fowke JH, Fahey JW, Stephenson KK, Hebert JR. Using isothiocyanate excretion as a biological marker of Brassica vegetable consumption in epidemiological studies: evaluating the sources of variability. *Public Health Nutr*. 2001 Jun;4(3):837-846.

183. Siddhanti SR, King MW, Tove SB. Influence of dietary fat on factors in serum that regulate thyroid cell metabolism. *N Nutr*. 1990 Nov;120(11):1297-1304.

184. Martino E, Buratti L, Bartalena L, et al. High prevalence of subacute thyroiditis during summer season in Italy. *J Endocrinol Invest*. 1987 Jun;10(3):321-323.

185. Jereczek-Fossa BA, Alterio D, Jassem J, Gibelli B, Tradati N, Orecchia R. Radiotherapy-induced thyroid disorders. *Cancer Treat Rev*. 2004 Jun;30(4):369-384.

186. van Santen HM, Vulsma T, Dijkgraaf MG, et al. No damaging effect of chemotherapy in addition to radiotherapy on the thyroid axis in young adult survivors of childhood cancer. *J Clin Endocrinol Metab*. 2003 Aug;88(8):3657-3663.

187. Dayan CM, Daniels GH. Chronic autoimmune thyroiditis. *N Engl J Med*. 1996 Jul 11;335(2):99-107.

188. Pearce EN, Farwell AP, Braverman LE. Thyroiditis. *N Engl J Med*. 2003 Jun 26;348(26):2646-2655.

189. Thieblemont C, Mayer A, Dumontet C, et al. Primary thyroid lymphoma is a heterogeneous disease. *J Clin Endocrinol Metab*. 2002 Jan;87(1):105-111.

190. Prasad ML, Huang Y, Pellegata NS, de la Chapelle A, Kloos RT. Hashimoto's thyroiditis with papillary thyroid carcinoma (PTC)-like nuclear alterations express molecular markers of PTC. *Histopathology*. 2004 Jul;45(1):39-46.

191. American Thyroid Association. Postpartum Thyroiditis. http://www.thyroid.org/wp-content/uploads/patients/brochures/Postpartum_Thyroiditis_brochure.pdf. Published 2012. Accessed December 27, 2013.

192. Stagnaro-Green A. Clinical review 152: Postpartum thyroiditis. *J Clin Endocrinol Metab*. 2002 Sep;87(9):4042-4047.

193. Premawardhana LD, Parkes AB, Ammari F, et al. Postpartum thyroiditis and long-term thyroid status: prognostic influence of thyroid peroxidase antibodies and ultrasound echogenicity. *J Clin Endocrinol Metab*. 2000 Jan;85(1):71-75.

194. Muller AF, Drexhage HA, Berghout A. Postpartum thyroiditis and autoimmune thyroiditis in women of childbearing age: recent insights and consequences for antenatal and postnatal care. *Endocr Rev*. 2001 Oct;22(5):605-630.

195. Martino E, Bartalena L, Bogazzi F, Braverman LE. The effects of amiodarone on the thyroid. *Endocr Rev*. 2001 Apr;22(2):240-254.

196. Carella C, Mazziotti G, Amato G, Braverman LE, Roti E. Clinical review 169: Interferon-alpha-related thyroid disease: pathophysiological, epidemiological, and clinical aspects. J Clin Endocrinol Metab. 2004 Aug;89(8):3656-3661.

197. Schuppert F, Rambusch E, Kirchner H, Atzpodien J, Kohn LD, von zur Mühlen A. Patients treated with interferon-alpha, interferon-beta, and interleukin-2 have a different thyroid autoantibody pattern than patients suffering from endogenous autoimmune thyroid disease. *Thyroid*. 1997 Dec;7(6):837-842.

198. Miller KK, Daniels GH. Association between lithium use and thyrotoxicosis caused by silent thyroiditis. *Clin Endocrinol* (Oxf). 2001 Oct;55(4):501-508.

199. Dang AH, Hershman JM. Lithium-associated thyroiditis. *Endocr Pract*. 2002 May-Jun;8(3):232-236.

200. Yale School of Medicine. Papillary Thyroid Cancer. http://www.yalesurgery.org/endocrine/carepeople/thyroid/papillary.aspx#page1. Accessed January 28, 2014.

201. Endocrineweb. Thyroid Cancer: Thyroid Cancer Symptoms, Diagnosis, and Treatments. http://www.endocrineweb.com/conditions/thyroid-cancer/thyroid-cancer. Accessed January 28, 2014.

202. CHEMIE.DE. Thyroid cancer. http://www.bionity.com/en/encyclopedia/Thyroid_cancer.html. Accessed January 28, 2014.

203. Hunt PJ, Marshall SE, Weetman AP, Bell JI, Wass JA, Welsh KI. Cytokine gene polymorphisms in autoimmune thyroid disease. *J Clin Endocrinol Metab*. 2000 May;85(5):1984-1988.

204. Collin P, Salmi J, Hällström O, Reunala T, Pasternack A. Autoimmune thyroid disorders and coeliac disease. *Eur J Endocrinol*. 1994 Feb;130(2):137-140.

205. Ahmad J, Ahmned F, Siddiqui MA, et al. Inflammatory markers, insulin resistance and carotid intima-media thickness in North-Indian type 2 diabetic subjects. *J Assoc Physicians India*. 2007 Oct;55:693-699.

206. Fernandez-Soto L, Gonzalez A, Lobón JA, Lopez JA, Peterson CM, Escobar-Jiménez F. Thyroid peroxidase autoantibodies predict poor metabolic control and need for thyroid treatment in pregnant IDDM women. *Diabetes Care*. 1997 Oct;20(10):1524-1528.

207. Drugarin D, Negru S, Koreck A, Zosin I, Cristea C. The pattern of a T(H)1 cytokine in autoimmune thyroiditis. *Immunol Lett*. 2000 Feb 1;71(2):73-77.

208. Smith EA, Frankenburg EP, Goldstein SA, et al. Effects of long-term administration of vitamin D3 analogs to mice. *J Endocrinol*. 2000 Apr;165(1):163-172.

209. Holick MF. Sunlight and vitamin D for bone health and prevention of autoimmune diseases, cancers, and cardiovascular disease. *Am J Clin Nutr*. 2004 Dec;80(6 Suppl):1678S-1688S.

210. Cantorna MT, Zhu Y, Froicu M, Wittke A. Vitamin D status, 1,25-dihydroxyvitamin D3, and the immune system. *Am J Clin Nutr*. 2004 Dec;80(6 Suppl):1717S-20S.

211. Deluca HF, Cantorna MT. Vitamin D: its role and uses in immunology. *FASEB J*. 2001 Dec;15(14):2579-2585.

212. Lin WY, Wan L, Tsai CH, Chen RH, Lee CC, Tsai FJ. Vitamin D receptor gene polymorphisms are associated with risk of Hashimoto's thyroiditis in Chinese patients in Taiwan. *J Clin Lab Anal*. 2006;20(3):109-112.

213. Koutkia P, Lu Z, Chen TC, Holick MF. Treatment of vitamin D deficiency due to Crohn's disease with tanning bed ultraviolet B radiation. *Gastroenterology*. 2001 Dec;121(6):1485-1488.

214. Vestergaard P. Prevalence and pathogenesis of osteoporosis in patients with inflammatory bowel disease. *Minerva Med*. 2004 Dec;95(6):469-480.

215. Wortsman J, Matsuoka LY, Chen TC, Lu Z, Holick MF. Decreased bioavailability of vitamin D in obesity. Am J Clin Nutr. 2000 Sep;72(3):690-693.

216. Holick MF. Photosynthesis of vitamin D in the skin: effect of environmental and life-style variables. *Fed Proc*. 1987 Apr;46(5):1876-1882.

217. Atli T, Gullu S, Uysal AR, Erdogan G. The prevalence of Vitamin D deficiency and effects of ultraviolet light on Vitamin D levels in elderly Turkish population. *Arch Gerontol Geriatr*. 2005 Jan-Feb;40(1):53-60.

218. Akçay MN, Akçay G. The presence of the antigliadin antibodies in autoimmune thyroid diseases. *Hepatogastroenterology*. 2003 Dec;50 Suppl 2:cclxxix-cclxxx.

219. Sategna-Guidetti C, Bruno M, Mazza E, et al. Autoimmune thyroid diseases and coeliac disease. *Eur J Gastroenterol Hepatol*. 1998 Nov;10(11):927-931.

220. Hakanen M, Luotola K, Salmi J, Laippala P, Kaukinen K, Collin P. Clinical and subclinical autoimmune thyroid disease in adult celiac disease. *Dig Dis Sci*. 2001 Dec;46(12):2631-2635.

221. Spadaccino AC, Basso D, Chiarelli S, et al. Celiac disease in North Italian patients with autoimmune thyroid diseases. *Autoimmunity*. 2008 Feb;41(1):116-121.

222. Hadithi M, de Boer H, Meijer JW, et al. Coeliac disease in Dutch patients with Hashimoto's thyroiditis and vice versa. *World J Gastroenterol*. 2007 Mar 21;13(11):1715-1722.

223. Silva CM, Souza MV. Autoimmune hypothyroidism nonresponsive to high doses of levothyroxine and severe hypocalcemia. *Arq Bras Endocrinol Metabol*. 2005 Aug;49(4):599-603.

224. Jiskra J, Límanová Z, Vanícková Z, Kocna P. IgA and IgG antigliadin, IgA anti-tissue transglutaminase and antiendomysial antibodies in patients with autoimmune thyroid diseases and their relationship to thyroidal replacement therapy. *Physiol Res*. 2003;52(1):79-88.

225. d'Estève-Bonetti L, Bennet AP, Malet D, Hoff M, Louvet JP, Caron P. Gluten-induced enteropathy (coeliac disease) revealed by resistance to treatment with levothyroxine and alfacalcidol in a sixty-eight-year-old patient: a case report. Thyroid. 2002 Jul;12(7):633-636.

226. Mainardi E, Montanelli A, Dotti M, Nano R, Moscato G. Thyroid-related autoantibodies and celiac disease: a role for a gluten-free diet? *J Clin Gastroenterol*. 2002 Sep;35(3):245-248.

227. Amino N, Hidaka Y, Takano T, Tatsumi KI, Izumi Y, Nakata Y. Possible induction of Graves' disease and painless thyroiditis by gonadotropin-releasing hormone analogues. *Thyroid*. 2003 Aug;13(8):815-818.

228. Ishii K, Hayashi A, Tamaoka A, Mizusawa H, Shoji S. A case of Hashimoto's encephalopathy with a relapsing course related to menstrual cycle. *Rinsho Shinkeigaku*. 1993 Sep;33(9):995-997.

229. Glinoer D, Riahi M, Grün JP, Kinthaert J. Risk of subclinical hypothyroidism in pregnant women with asymptomatic autoimmune thyroid disorders. *J Clin Endocrinol Metab*. 1994 Jul;79(1):197-204.

230. Guan HX, Li CY, Li YS, et al. Thyroid function and thyroid autoimmunity at the late pregnancy: data from 664 pregnant women. *Zhonghua Fu Chan Ke Za Zhi*. 2006 Aug;41(8):529-532.

231. Massoudi MS, Meilahn EN, Orchard TJ, et al. Prevalence of thyroid antibodies among healthy middle-aged women. Findings from the thyroid study in healthy women. *Ann Epicemiol*. 1995 May;5(3):229-233.

232. Surks MI, Sievert R. Drugs and thyroid function. *N Engl J Med*. 1995 Dec 21;333(25):1688-1694.

233. Weetman AP. Graves' disease. *N Engl J Med*. 2000 Oct 26;343(17):1236-1248.

234. Teng W, Shan Z, Teng X, et al. Effect of iodine intake on thyroid diseases in China. *N Engl J Med*. 2006 Jun 29;354(26):2783-2793.

235. Mazziotti G, Premawardhana LD, Parkes AB, et al. Evolution of thyroid autoimmunity during iodine prophylaxis--the Sri Lankan experience. *Eur J Endocrinol*. 2003 Aug;149(2):103-110.

236. Bastemir M, Emral R, Erdogan G, Gullu S. High prevalence of thyroid dysfunction and autoimmune thyroiditis in adolescents after elimination of iodine deficiency in the Eastern Black Sea Region of Turkey. *Thyroid*. 2006 Dec;16(12):1265-1271.

237. Zhao SJ, Sun FJ, Tian EJ, Chen ZP. Experimental study on effects of iodine deficiency and excess on thyroid autoimmunity. *Zhonghua Yu Fang Yi Xue Za Zhi*. 2006 Jan;40(1):18-20.

238. McLachlan SM, Rapoport B. Why measure thyroglobulin autoantibodies rather than thyroid peroxidase autoantibodies? *Thyroid*. 2004 Jul;14(7):510-520.

239. Camargo RY, Tomimori EK, Neves SC, et al. Thyroid and the environment: exposure to excessive nutritional iodine increases the prevalence of thyroid disorders in Sao Paulo, Brazil. *Eur J Endocrinol*. 2008 Sep;159(3):293-299.

240. Zonenberg A, Telejko B, Szelachowska M, Modzelewska A, Nikołajuk A, Górska M. Markers of endothelial dysfunction in patients with iodine induced hyperthyroidism. *Endokrynol Pol*. 2006 May-Jun;57(3):210-217.

241. Chong W, Shi XG, Teng WP, et al. Multifactor analysis of relationship between the biological exposure to iodine and hypothyroidism. *Zhonghua Yi Xue Za Zhi*. 2004 Jul 17;84(14):1171-1174.

242. Zois C, Stavrou I, Kalogera C, et al. High prevalence of autoimmune thyroiditis in schoolchildren after elimination of iodine deficiency in northwestern Greece. *Thyroid*. 2003 May;13(5):485-489.

243. Markou KB, Georgopoulos NA, Makri M, et al. Improvement of iodine deficiency after iodine supplementation in schoolchildren of Azerbaijan was accompanied by hypo and hyperthyrotropinemia and increased title of thyroid autoantibodies. J Endocrinol Invest. 2003;26(2 Suppl):43-48.

244. Yoon SJ, Choi SR, Kim DM, et al. The effect of iodine restriction on thyroid function in patients with hypothyroidism due to Hashimoto's thyroiditis. *Yonsei Med J*. 2003 Apr 30;44(2):227-235.

245. Sterzl I, Prochazkova J, Hrda P, Matucha P, Bartova J, Stejskal V. Removal of dental amalgam decreases anti-TPO and anti-Tg autoantibodies in patients with autoimmune thyroiditis. *Neuro Endocrinol Lett*. 2006 Dec;27 Suppl 1:25-30.

246. Tomer Y, Davies TF. Infection, thyroid disease, and autoimmunity. *Endocr Rev*. 1993 Feb;14(1):107-120.

247. Rasmussen AK, Feldt-Rasmussen UF. Autoimmune thyroiditis—an infectious disease? *Ugeskr Laeger*. 2002 Dec 9;164(50):5911-5915.

248. Desailloud R, Hober D. Viruses and thyroiditis: an update. *Virol J*. 2009 Jan 12;6:5.

249. Edlich RF, Winters KL, Long WB 3rd, Gubler KD. Rubella and congenital rubella (German measles). *J Long Term Eff Med Implants*. 2005;15(3):319-328.

250. Best JM. Rubella. Semin Fetal Neonatal Med. 2007 Jun;12(3):182-192.

251. Stegmann BJ, Carey JC. TORCH Infections. Toxoplasmosis, Other (syphilis, varicella-zoster, parvovirus B19), Rubella, Cytomegalovirus (CMV), and Herpes infections. *Curr Womens Health Rep*. 2002 Aug;2(4):253-258.

252. Lindberg B, Ahlfors K, Carlsson A, et al. Previous exposure to measles, mumps, and rubella—but not vaccination during adolescence—correlates to the prevalence of pancreatic and thyroid autoantibodies. *Pediatrics*. 1999 Jul;104(1):e12.

253. Miyai K, Mizuta H, Kurimura T, Nose O, Tsuruhara T. Rubella antibody in neonatal hypothyroidism due to thyroid dysgenesis. *Early Hum Dev*. 1981 May;5(2):215-216.

254. Clarke WL, Shaver KA, Bright GM, Rogol AD, Nance WE. Autoimmunity in congenital rubella syndrome. *J Pediatr*. 1984 Mar;104(3):370-373.

255. Nakamura S, Kosaka J, Sugimoto M, Watanabe H, Shima H, Takuno H. Silent thyroiditis following rubella. *Endocrinol Jpn*. 1990 Feb;37(1):79-85.

256. Yates JL, Warren N, Sugden B. Stable replication of plasmids derived from Epstein-Barr virus in various mammalian cells. *Nature*. 1985 Feb 28-Mar 6;313(6005):812-815.

257. Volta C, Carano N, Street ME, Bernasconi S. Atypical subacute thyroiditis caused by Epstein-Barr virus infection in a three-year-old girl. *Thyroid*. 2005 Oct;15(10):1189-1191.

258. Benvenga S, Trimarchi F. Thyroid hormone autoantibodies in Hashimoto's thyroiditis: often transient but also increasingly frequent. *Thyroid*. 2003 Oct;13(10):995-996.

259. Espino Montoro A, Medina Pérez M, González Martín MC, Asencio Marchante R, López Chozas JM. Subacute thyroiditis associated with positive antibodies to the Epstein-Barr virus. *An Med Interna*. 2000 Oct;17(10):546-548.

260. Thomas D, Karachaliou F, Kallergi K, et al. Herpes virus antibodies seroprevalence in children with autoimmune thyroid disease. *Endocrine*. 2008 Apr;33(2):171-175.

261. Nakamura M, Burastero SE, Ueki Y, Larrick JW, Notkins AL, Casali P. Probing the normal and autoimmune B cell repertoire with Epstein-Barr virus. Frequency of B cells producing monoreactive high affinity autoantibodies in patients with Hashimoto's disease and systemic lupus erythematosus. *J Immunol*. 1988 Dec 15;141(12):4165-4172.

262. Shimon I, Pariente C, Shlomo-David J, Grossman Z, Sack J. Transient elevation of triiodothyronine caused by triiodothyronine autoantibody associated with acute Epstein-Barr-virus infection. *Thyroid*. 2003 Feb;13(2):211-215.

263. Tomoyose T, Komiya I, Takara M, et al. Cytotoxic T-lymphocyte antigen-4 gene polymorphisms and human T-cell lymphotrophic virus-1 infection: their associations with Hashimoto's thyroiditis in Japanese patients. *Thyroid*. 2002 Aug;12(8):673-677.

264. Wick G, Trieb K, Aguzzi A, Recheis H, Anderl H, Grubeck-Loebenstein B. Possible role of human foamy virus in Graves' disease. *Intervirology*. 1993;35(1-4):101-107.

265. Kawai H, Mitsui T, Yokoi K, et al. Evidence of HTLV-I in thyroid tissue in an HTLV-I carrier with Hashimoto's thyroiditis. *J Mol Med* (Berl). 1996 May;74(5):275-278.

266. Kawai H, Saito M, Takagi M, et al. Hashimoto's thyroiditis in HTLV-I carriers. *Intern Med*. 1992 Oct;31(10):1213-1216.

267. Schattner A, Rager-Zisman B. Virus-induced autoimmunity. *Rev Infect Dis*. 1990 Mar-Apr;12(2):204-222.

268. Taniguchi Y, Yorioka N, Katsutani M, et al. Hemophagocytic syndrome in a patient with Hashimoto's thyroiditis and membranous nephritis. *Nephron*. 1999 Feb;81(2):246-247.

269. Prummel MF, Laurberg P. Interferon-alpha and autoimmune thyroid disease. *Thyroid*. 2003 Jun;13(6):547-551.

270. Stancek D, Stanceková-Gressnerová M, Janotka M, Hnilica P, Oravec D. Isolation and some serological and epidemiological data on the viruses recovered from patients with subacute thyroiditis de Quervain. *Med Microbiol Immunol*. 1975;161(2):133-144.

271. Kraemer MH, Donadi EA, Tambascia MA, Magna LA, Prigenzi LS. Relationship between HLA antigens and infectious agents in contributing towards the development of Graves' disease. *Immunol Invest*. 1998 Jan-Feb;27(1-2):17-29.

272. Aleksandrova GF, Efuni SS, Lavrova IK, Savinov AP, Bel'chenko LV. The role of Coxsackie viruses in the pathogenesis of diffuse toxic goiter. *Probl Endokrinol* (Mosk). 1988 Nov-Dec;34(6):18-20.

273. Lehmann HW, Lutterbüse N, Plentz A, et al. Association of parvovirus B19 infection and Hashimoto's thyroiditis in children. *Viral Immunol*. 2008 Sep;21(3):379-383.

274. Arscott P, Rosen ED, Koenig RJ, et al. Immunoreactivity to Yersinia enterocolitica antigens in patients with autoimmune thyroid disease. *J Clin Endocrinol Metab*. 1992 Jul;75(1):295-300.

275. Strieder TG, Wenzel BE, Prummel MF, Tijssen JG, Wiersinga WM. Increased prevalence of antibodies to enteropathogenic Yersinia enterocolitica virulence proteins in relatives of patients with autoimmune thyroid disease. *Clin Exp Immunol*. 2003 May;132(2):278-82.

276. Guan H, Li C, Li Y, et al. High iodine intake is a risk factor of post-partum thyroiditis: result of a survey from Shenyang, China. *J Endocrinol Invest*. 2005 Nov;28(10):876-881.

277. Pedersen IB, Knudsen N, Jørgensen T, Perrild H, Ovesen L, Laurberg P. Thyroid peroxidase and thyroglobulin autoantibodies in a large survey of populations with mild and moderate iodine deficiency. *Clin Endocrinol* (Oxf). 2003 Jan;58(1):36-42.

278. Olivieri O, Girelli D, Azzini M, et al. Low selenium status in the elderly influences thyroid hormones. *Clin Sci* (Lond). 1995 Dec;89(6):637-642.

279. Kralik A, Eder K, Kirchgessner M. Influence of zinc and selenium deficiency on parameters relating to thyroid hormone metabolism. *Horm Metab Res*. 1996 May;28(5):223-226.

280. Calomme M, Vanderpas J, François B, et al. Effects of selenium supplementation on thyroid hormone metabolism in phenylketonuria subjects on a phenylalanine restricted diet. *Biol Trace Elem Res*. 1995 Jan-Mar;47(1-3):349-353.

281. Lehmann C, Egerer K, Weber M, et al. Effect of selenium administration on various laboratory parameters of patients at risk for sepsis syndrome. *Med Klin* (Munich). 1997 Sep 15;92 Suppl 3:14-16.

282. Kauf E, Dawczynski H, Jahreis G, Janitzky E, Winnefeld K. Sodium selenite therapy and thyroid-hormone status in cystic fibrosis and congenital hypothyroidism. *Biol Trace Elem Res*. 1994 Mar;40(3):247-253.

283. Olivieri O, Girelli D, Stanzial AM, Rossi L, Bassi A, Corrocher R. Selenium, zinc, and thyroid hormones in healthy subjects: low T3/T4 ratio in the elderly is related to impaired selenium status. *Biol Trace Elem Res*. 1996 Jan;51(1):31-41.

284. Zhu Z, Kimura M, Itokawa Y. Iodothyronine deiodinase activity in methionine-deficient rats fed selenium-deficient or selenium-sufficient diets. *Biol Trace Elem Res*. 1995 May;48(2):197-213.

285. Fujimoto S, Indo Y, Higashi A, Matsuda I, Kashiwabara N, Nakashima I. Conversion of thyroxine into tri-iodothyronine in zinc deficient rat liver. *J Pediatr Gastroenterol Nutr*. 1986 Sep-Oct;5(5):799-805.

286. Nishiyama S, Futagoishi-Suginohara Y, Matsukura M. Zinc supplementation alters thyroid hormone metabolism in disabled patients with zinc deficiency. *J Am Coll Nutr*.1994 Feb;13(1):62-67.

287. Licastro F, Mocchegiani E, Zannotti M, Arena G, Masi M, Fabris N. Zinc affects the metabolism of thyroid hormones in children with Down's syndrome: normalization of thyroid stimulating hormone and of reversal triiodothyronine plasmic levels by dietary zinc supplementation. *Int J Neurosci*. 1992 Jul-Aug;65(1-4):259-268.

288. Limpach A, Dalton M, Miles R, Gadson P. Homocysteine inhibits retinoic acid synthesis: a mechanism for homocysteine-induced congenital defects. *Exp Cell Res*. 2000 Oct 10;260(1):166-174.

Additional Case Studies

Case 41

TSH: Elevated

Total T4: Normal Range

T3 Uptake: Elevated

Total T3: Elevated

Free T3: Elevated

Free T4: Elevated

TBG: Depressed

TPO Ab: Positive

Antithyroglobulin Ab: Negative

Case 42

TSH: Low

Total T4: Low

T3 Uptake: Normal Range

Total T3: Low Normal

Free T3: Normal Range

Free T4: Normal Range

TBG: Normal Range

TPO Ab: Negative

Antithyroglobulin Ab: Negative

Case 43

TSH: Normal Range

Total T4: Normal Range

T3 Uptake: Normal Range

Total T3: Normal Range

Free T3: Normal Range

Free T4: Normal Range

TBG: Normal Range

TPO Ab: Positive

Antithyroglobulin Ab: Negative

Case 44

TSH: Normal Range

Total T4: Normal Range

T3 Uptake: Low

Total T3: Normal Range

Free T3: Low

Free T4: Low

TBG: Elevated

TPO Ab: Negative

Antithyroglobulin Ab: Negative

Case 45

TSH: Elevated

Total T4: Normal Range

T3 Uptake: Normal Range

Total T3: Normal Range

Free T3: Normal Range

Free T4: Normal Range

TBG: Normal Range

TPO Ab: Negative

Antithyroglobulin Ab: Negative

Case 46

TSH: Severely Depressed

Total T4: Normal Range

T3 Uptake: Normal Range

Total T3: Normal Range

Free T3: Elevated

Free T4: Elevated

TBG: Normal Range

TPO Ab: Positive

Antithyroglobulin Ab: Positive

Case 47

TSH: Elevated

Total T4: Normal Range

T3 Uptake: Normal Range

Total T3: Normal Range

Free T3: Normal Range

Free T4: Normal Range

TBG: Normal Range

TPO Ab: Positive

Antithyroglobulin Ab: Negative

Case 48

TSH: Normal Range

Total T4: Normal Range

T3 Uptake: Elevated

Total T3: Normal Range

Free T3: Elevated

Free T4: Elevated

TBG: Depressed

TPO Ab: Negative

Antithyroglobulin Ab: Negative

Case 49

TSH: Normal Range

Total T4: Normal Range

T3 Uptake: Normal Range

Total T3: Normal Range

Free T3: Normal Range

Free T4: Normal Range

TBG: Normal Range

TPO Ab: Negative

Antithyroglobulin Ab: Negative

Case 50

TSH: Depressed

Total T4: Normal Range

T3 Uptake: Normal Range

Total T3: Low

Free T3: Low

Free T4: Normal Range

TBG: Normal Range

TPO Ab: Negative

Antithyroglobulin Ab: Negative

Case 51

TSH: Elevated

Total T4: Normal Range

T3 Uptake: Normal Range

Total T3: Low

Free T3: Low

Free T4: Normal Range

TBG: Normal Range

TPO Ab: Negative

Antithyroglobulin Ab: Negative

Case 52

TSH: Elevated

TT4: Normal Range

T3 Uptake: Normal Range

Total T3: Low

Free T3: Low

Free T4: Normal Range

TBG: Normal Range

TPO Ab: Negative

Antithyroglobulin Ab: Negative

Case 53

TSH: Elevated

Total T4: Normal Range

T3 Uptake: Depressed

Total T3: Low

Free T3: Low

Free T4: Low

TBG: Elevated

TPO Ab: Positive

Antithyroglobulin Ab: Positive

Case 54

TSH: Depressed

Total T4: Normal Range

T3 Uptake: Normal Range

Total T3: Normal Range

Free T3: Normal Range

Free T4: Normal Range

TBG: Normal Range

TPO Ab: Negative

Antithyroglobulin Ab: Negative

Case 55

TSH: Normal Range

Total T4: Normal Range

T3 Uptake: Normal Range

Total T3: Normal Range

Free T3: Elevated

Free T4: Normal Range

TBG: Negative

TPO Ab: Positive

Antithyroglobulin Ab: Positive

Case 56

TSH: Elevated

Total T4: Normal Range

T3 Uptake: Elevated

Total T3: Normal Range

Free T3: Elevated

Free T4: Elevated

TBG: Depressed

TPO Ab: Negative

Antithyroglobulin Ab: Negative

Case 57

TSH: Elevated

Total T4: Normal Range

T3 Uptake: Normal Range

Total T3: Normal Range

Free T3: Normal Range

Free T4: Normal Range

TBG: Normal Range

TPO Ab: Negative

Antithyroglobulin Ab: Negative

Case 58

TSH: Elevated

Total T4: Normal Range

T3 Uptake: Elevated

Total T3: Normal Range

Free T3: Elevated

Free T4: Elevated

TBG: Depressed

TPO Ab: Positive

Antithyroglobulin Ab: Positive

Case 59

TSH: Depressed

Total T4: Depressed

T3 Uptake: Normal Range

Total T3: Depressed

Free T3: Depressed

Free T4: Depressed

TBG: Normal Range

TPO Ab: Negative

Antithyroglobulin Ab: Positive

Case 60

TSH: Depressed

Total T4: Elevated

T3 Uptake: Normal Range

Total T3: Elevated

Free T3: Elevated

Free T4: Elevated

TBG: Normal Range

TPO Ab: Positive

Antithyroglobulin Ab: Positive

Case 61

TSH: Depressed

TT4: Elevated

T3 Uptake: Normal Range

TT3: Elevated

FT3: Elevated

FT4: Elevated

TBG: Normal Range

TPO Ab: Positive

Antithyroglobulin Ab: Positive

TSI Ab: Positive

TSH-Receptor Ab: Positive

Case 62

TSH: Depressed

Total T4: Elevated

T3 Uptake: Normal Range

Total T3: Elevated

Free T3: Elevated

Free T4: Elevated

TBG: Normal Range

TPO Ab: Positive

Antithyroglobulin Ab: Positive

TSI Ab: Negative

TSH-Receptor Ab: Negative

Case 63

TSH: Normal Range

Total T4: Normal Range

T3 Uptake: Normal Range

Total T3: Depressed

Reverse T3: Elevated

Free T3: Depressed

Free T4: Normal Range

TBG: Normal Range

TPO Ab: Negative

Antithyroglobulin Ab: Negative

Case 64

TSH: Depressed

Total T4: Normal Range

T3 Uptake: Normal Range

Total T3: Depressed

Reverse T3: Elevated

Free T3: Depressed

Free T4: Normal Range

TBG: Normal Range

TPO Ab: Normal Range

Antithyroglobulin Ab: Negative

Case 65

TSH: Depressed

Total T4: Normal Range

T3 Uptake: Normal Range

Total T3: Depressed

Reverse T3: Elevated

Free T3: Depressed

Free T4: Normal Range

TBG: Normal Range

TPO Ab: Positive

Antithyroglobulin Ab: Negative

Case 66

TSH: Depressed

Total T4: Depressed

T3 Uptake: Depressed

Total T3: Normal Range

Free T3: Depressed

Free T4: Depressed

TBG: Elevated

TPO Ab: Negative

Antithyroglobulin Ab: Negative

Case 67

TSH: Depressed

Total T4: Depressed

T3 Uptake: Normal Range

Total T3: Normal Range

Free T3: Normal Range

Free T4: Normal Range

TBG: Normal Range

TPO Ab: Negative

Antithyroglobulin Ab: Negative

Case 68

TSH: Severely Depressed

Total T4: Normal Range

T3 Uptake: Normal Range

Total T3: Normal Range

Free T3: Normal Range

Free T4: Normal Range

TBG: Normal Range

TPO Ab: Negative

Antithyroglobulin Ab: Positive

Case 69

TSH: Depressed

Total T4: Depressed

T3 Uptake: Depressed

Total T3: Normal Range

Free T3: Depressed

Free T4: Depressed

TBG: Elevated

TPO Ab: Positive

Antithyroglobulin Ab: Negative

Case 70

TSH: Normal Range

Total T4: Normal Range

T3 Uptake: Normal Range

Total T3: Depressed

Free T3: Depressed

Reverse T3: Elevated

Free T4: Normal Range

TBG: Normal Range

TPO Ab: Positive

Antithyroglobulin Ab: Positive

Case 71

TSH: Normal Range

Total T4: Normal Range

T3 Uptake: Depressed

Total T3: Depressed

Free T3: Depressed

Reverse T3: Elevated

Free T4: Depressed

TBG: Elevated

TPO Ab: Positive

Antithyroglobulin Ab: Positive

Case 72

TSH: Normal Range

Total T4: Normal Range

T3 Uptake: Normal Range

Total T3: Depressed

Free T3: Depressed

Reverse T3: Normal Range

Free T4: Normal Range

TBG: Normal Range

TPO Ab: Negative

Antithyroglobulin Ab: Negative

Case 73

TSH: Elevated

Total T4: Normal Range

T3 Uptake: Depressed

Total T3: Normal Range

Free T3: Normal Range

FT4: Normal Range

TBG: Elevated

TPO Ab: Positive

Antithyroglobulin Ab: Negative

Case 74

TSH: Normal Range

Total T4: Normal Range

T3 Uptake: Depressed

Total T3:Normal Range

Free T3: Depressed

Free T4: Depressed

Tg Ab: Strongly Positive

TPO Ab: Strongly Positive

Antithyroglobulin Ab: Strongly Positive

Case 75

TSH: Severely Depressed

Total T4: Elevated

T3 Uptake: Depressed

Total T3:Elevated

Free T3: Normal Range

Free T4: Normal Range

TG: Strongly Positive

TPO Ab: Strongly Positive

Antithyroglobulin Ab: Strongly Positive